

R2 Newton's second law

Name: _____

Class: _____

Date: _____

Time: **434 minutes**

Marks: **363 marks**

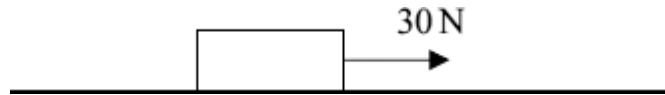
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Q1.

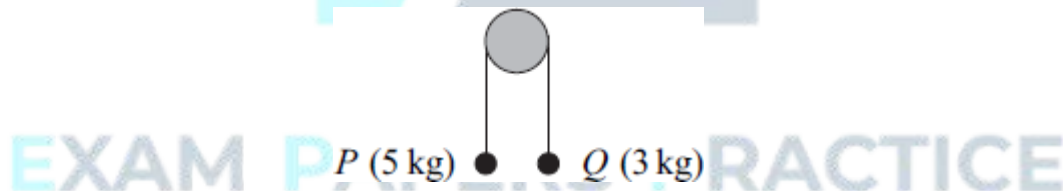
A wooden block, of mass 4 kg, is placed on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.3. A horizontal force, of magnitude 30 newtons, acts on the block and causes it to accelerate.



- (a) Draw a diagram to show all the forces acting on the block. (1)
 - (b) Calculate the magnitude of the normal reaction force acting on the block. (1)
 - (c) Find the magnitude of the friction force acting on the block. (2)
 - (d) Find the acceleration of the block. (3)
- (Total 7 marks)**

Q2.

Two particles, P and Q , are connected by a string that passes over a fixed smooth peg, as shown in the diagram. The mass of P is 5 kg and the mass of Q is 3 kg.



The particles are released from rest in the position shown.

- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.45 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) State **two** modelling assumptions that you have made about the string. (2)
- (d) Particle P hits the floor when it has moved 0.196 metres and Q has not reached the peg.
 - (i) Find the time that it takes P to reach the floor. (3)

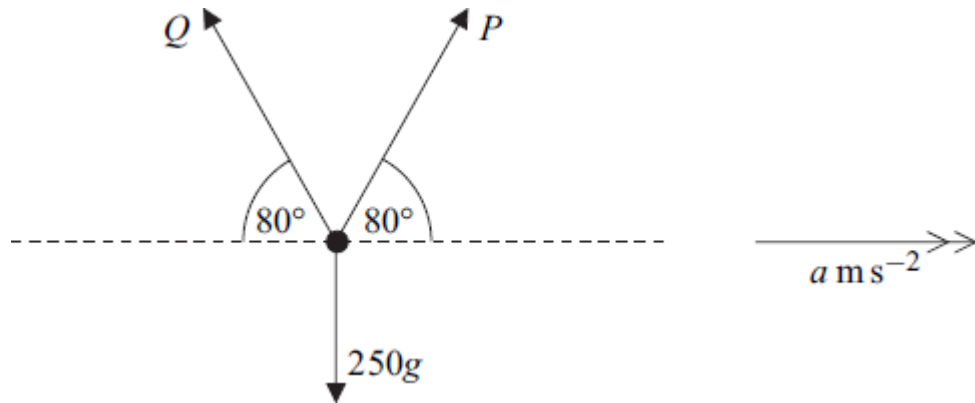
- (ii) Find the speed of P when it hits the floor.

(2)

(Total 14 marks)

Q3.

Three forces act in a vertical plane on an object of mass 250 kg, as shown in the diagram.



The two forces P newtons and Q newtons each act at 80° to the horizontal. The object accelerates horizontally at $a \text{ m s}^{-2}$ under the action of these forces.

- (a) Show that

$$P = 125 \left(\frac{a}{\cos 80^\circ} + \frac{g}{\sin 80^\circ} \right) \quad (5)$$

- (b) Find the value of a for which Q is zero.

(3)

(Total 8 marks)

Q4.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the velocity of the particle, $\mathbf{v} \text{ m s}^{-1}$, is given by

$$\mathbf{v} = 4e^{-2t} \mathbf{i} + (6t - 3t^2) \mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t .

(3)

- (b) The mass of the particle is 5 kg.

- (i) Find an expression for the force $\mathbf{F}$ acting on the particle at time t .

(2)

- (ii) Find the magnitude of $\mathbf{F}$ when $t = 0$.

(2)

- (c) Find the value of t when $\mathbf{F}$ acts due west.

(2)

- (d) When $t = 0$, the particle is at the point with position vector $(6\mathbf{i} + 5\mathbf{j})$ m.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 14 marks)

Q5.

A car, of mass m kg, is moving along a straight horizontal road. At time t seconds, the car has speed v m s⁻¹. As the car moves, it experiences a resistance force of magnitude $2mv^{\frac{5}{4}}$ newtons. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -2v^{\frac{5}{4}}$$

(1)

- (b) The initial speed of the car is 16 m s⁻¹.

Show that

$$v = \left(\frac{2}{t+1} \right)^4$$

(5)

(Total 6 marks)

Q6.

A boat moves with constant acceleration, so that its velocity $\mathbf{v}$ m s⁻¹ at time t seconds is given by

$$\mathbf{v} = (4.5 - 0.9t)\mathbf{i} + (2 + 0.5t)\mathbf{j}$$

where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the velocity of the boat when $t = 0$.

(1)

- (b) Find the velocity of the boat when $t = 5$.

(1)

- (c) State the direction in which the boat is travelling when $t = 5$.

(1)

- (d) Find the acceleration of the boat.

(3)

- (e) The mass of the boat is 250 kg. Find the magnitude of the resultant force acting on the boat. (3)
- (f) Draw a diagram to show the direction of the resultant force on the boat and calculate the angle between this force and the unit vector $\mathbf{i}$. (4)

(Total 13 marks)

Q7.

A ball is released from rest at a height of 15 metres above ground level.

- (a) Find the speed of the ball when it hits the ground, assuming that no air resistance acts on the ball. (3)
- (b) In fact, air resistance does act on the ball. Assume that the air resistance force has a constant magnitude of 0.9 newtons. The ball has a mass of 0.5 kg.
- (i) Draw a diagram to show the forces acting on the ball, including the magnitudes of the forces acting. (1)
- (ii) Show that the acceleration of the ball is 8 m s^{-2} . (3)
- (iii) Find the speed at which the ball hits the ground. (2)
- (iv) Explain why the assumption that the air resistance force is constant may not be valid. (1)

(Total 10 marks)

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Q8.

The constant forces $\mathbf{F}_1 = (8\mathbf{i} + 12\mathbf{j})$ newtons and $\mathbf{F}_2 = (4\mathbf{i} - 4\mathbf{j})$ newtons act on a particle. No other forces act on the particle.

- (a) Find the resultant force acting on the particle. (2)
- (b) Given that the mass of the particle is 4 kg, show that the acceleration of the particle is $(3\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. (2)
- (c) At time t seconds, the velocity of the particle is $\mathbf{v} \text{ m s}^{-1}$.
- (i) When $t = 20$, $\mathbf{v} = 40\mathbf{i} + 32\mathbf{j}$.

Show that $\mathbf{v} = -20\mathbf{i} - 8\mathbf{j}$ when $t = 0$.

(3)

- (ii) Write down an expression for v at time t .

(1)

- (iii) Find the times when the speed of the particle is 8 m s^{-1} .

(6)

(Total 14 marks)

Q9.

A small train at an amusement park consists of an engine and two carriages connected to each other by light horizontal rods, as shown in the diagram.



The engine has mass 2000 kg and each carriage has mass 500 kg .

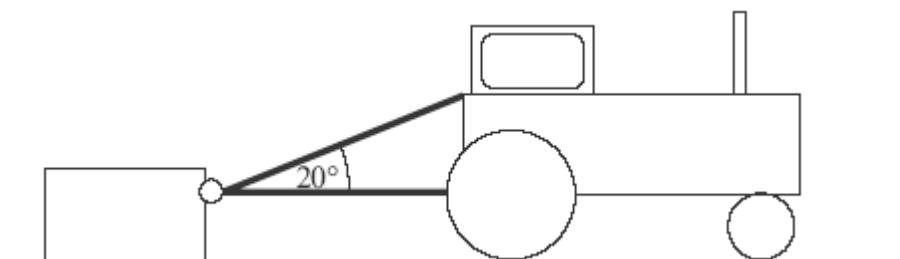
The train moves along a straight horizontal track. A resistance force of magnitude 400 newtons acts on the engine, and resistance forces of magnitude 300 newtons act on each carriage. The train is accelerating at 0.5 m s^{-2} .

- Draw a diagram to show the **horizontal** forces acting on Carriage 2. (1)
- Show that the magnitude of the force that the rod exerts on Carriage 2 is 550 newtons . (2)
- Find the magnitude of the force that the rod attached to the engine exerts on Carriage 1. (3)
- A forward driving force of magnitude $P \text{ newtons}$ acts on the engine. Find P . (3)

(Total 9 marks)

Q10.

A crate, of mass 200 kg , is initially at rest on a rough horizontal surface. A smooth ring is attached to the crate. A light inextensible rope is passed through the ring, and each end of the rope is attached to a tractor. The lower part of the rope is horizontal and the upper part is at an angle of 20° to the horizontal, as shown in the diagram.



When the tractor moves forward, the crate accelerates at 0.3 m s^{-2} . The coefficient of friction between the crate and the surface is 0.4.

Assume that the tension, T newtons, is the same in both parts of the rope.

- (a) Draw and label a diagram to show the forces acting on the crate. (2)
 - (b) Express the normal reaction between the surface and the crate in terms of T . (3)
 - (c) Find T . (5)
- (Total 10 marks)

Q11.

A golf ball, of mass m kg, is moving in a straight line across smooth horizontal ground. At time t seconds, the golf ball has speed $v \text{ m s}^{-1}$. As the golf ball moves, it experiences

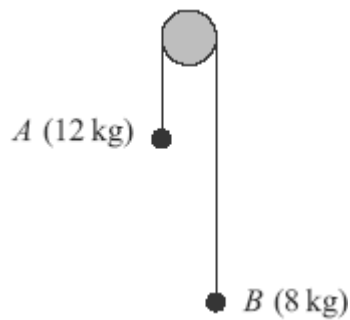
a resistance force of magnitude $0.2mv^{\frac{1}{2}}$ newtons until it comes to rest. No other horizontal force acts on the golf ball.

Model the golf ball as a particle.

- (a) Show that
$$\frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$
 (1)
 - (b) When $t = 0$, the speed of the golf ball is 16 m s^{-1} .
Show that $v = (4 - 0.1t)^2$. (5)
 - (c) Find the value of t when $v = 1$. (3)
 - (d) Find the distance travelled by the golf ball as its speed decreases from 16 m s^{-1} to 1 m s^{-1} . (4)
- (Total 13 marks)

Q12.

Two particles, A and B , have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.

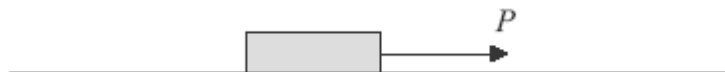


The particles are released from rest and move vertically. Assume that there is no air resistance.

- (a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2} . (5)
 - (b) Find the tension in the string. (2)
 - (c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.
 - (i) Find the speed of particle *A* when the string breaks. (2)
 - (ii) Find the speed of particle *A* when it hits the surface. (3)
- (Total 12 marks)

Q13.

A block, of mass 10 kg, is at rest on a rough horizontal surface, when a horizontal force, of magnitude P newtons, is applied to the block, as shown in the diagram.



The coefficient of friction between the block and the surface is 0.5.

- (a) Draw and label a diagram to show all the forces acting on the block. (1)
- (b) (i) Calculate the magnitude of the normal reaction force acting on the block. (1)
- (ii) Find the maximum possible magnitude of the friction force between the block and the surface. (1)
- (iii) Given that $P = 30$, state the magnitude of the friction force acting on the block.

(1)

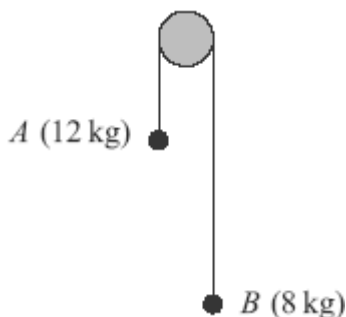
- (c) Given that $P = 80$, find the acceleration of the block.

(3)

(Total 7 marks)

Q14.

Two particles, A and B , have masses 12 kg and 8 kg respectively. They are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram.



The particles are released from rest and move vertically. Assume that there is no air resistance.

- (a) By forming two equations of motion, show that the magnitude of the acceleration of each particle is 1.96 m s^{-2} .

(5)

- (b) Find the tension in the string.

(2)

- (c) After the particles have been moving for 2 seconds, both particles are at a height of 4 metres above a horizontal surface. When the particles are in this position, the string breaks.

- (i) Find the speed of particle A when the string breaks.

(2)

- (ii) Find the speed of particle A when it hits the surface.

(3)

- (iii) Find the time that it takes for particle B to reach the surface after the string breaks.

Assume that particle B does not hit the peg.

(5)

(Total 17 marks)

Q15.

A particle, of mass 10 kg, moves on a smooth horizontal surface. A single horizontal force, $(9\mathbf{i} + 12\mathbf{j})$ newtons, acts on the particle.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the acceleration of the particle. (2)
- (b) At time t seconds, the velocity of the particle is $\mathbf{v} \text{ m s}^{-1}$. When $t = 0$, the velocity of the particle is $(2.2\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$ and the particle is at the origin.
- (i) Find the distance between the particle and the origin when $t = 5$. (4)
- (ii) Express $\mathbf{v}$ in terms of t . (2)
- (iii) Find t when the particle is travelling north-east. (3)
- (Total 11 marks)**

Q16.

A particle has mass 200 kg and moves on a smooth horizontal plane. A single

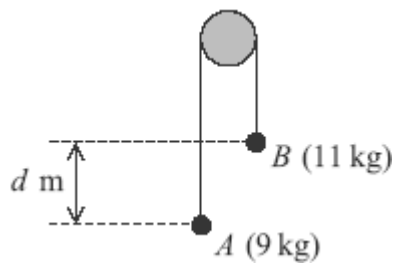
horizontal force, $\left(400 \cos\left(\frac{\pi}{2}t\right)\mathbf{i} + 600t^2\mathbf{j}\right)$ newtons, acts on the particle at time t seconds.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the acceleration of the particle at time t . (2)
- (b) When $t = 4$, the velocity of the particle is $(-3\mathbf{i} + 56\mathbf{j}) \text{ m s}^{-1}$.
Find the velocity of the particle at time t . (5)
- (c) Find t when the particle is moving due west. (3)
- (d) Find the speed of the particle when it is moving due west. (2)
- (Total 12 marks)**

Q17.

Two particles, A and B , are connected by a light inextensible string that passes over a smooth fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A .

Assume that no resistance forces act on the particles.

- (a) By forming an equation of motion for each of the particles A and B , show that the acceleration of each particle has magnitude 0.98 m s^{-2} .

(5)

- (b) When the particles have been moving for 0.5 seconds, they are at the same level.

- (i) Find the speed of the particles at this time.

(2)

- (ii) Find d .

(4)

(Total 11 marks)

Q18.

A box of mass 4 kg is held at rest on a plane inclined at an angle of 40° to the horizontal. The box is then released and slides down the plane.

- (a) A simple model assumes that the only forces acting on the box are its weight and the normal reaction from the plane. Show that, according to this simple model, the acceleration of the box would be 6.30 m s^{-2} , correct to three significant figures.

(3)

- (b) In fact, the box moves down the plane with constant acceleration and travels 0.9 metres in 0.6 seconds. By using this information, find the acceleration of the box.

(3)

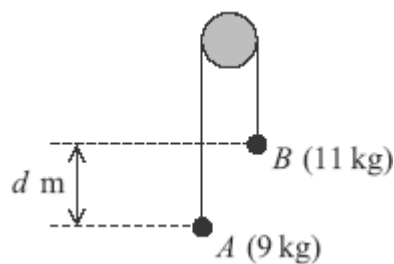
- (c) Explain why the answer to part (b) is less than the answer to part (a).

(1)

(Total 7 marks)

Q19.

Two particles, A and B , are connected by a string that passes over a fixed peg, as shown in the diagram. The mass of A is 9 kg and the mass of B is 11 kg.



The particles are released from rest in the position shown, where B is d metres higher than A .

The motion of the particles is to be modelled using simple assumptions.

- (a) State one assumption that should be made about the peg. (1)
- (b) State two assumptions that should be made about the string. (2)
- (c) By forming an equation of motion for each of the particles A and B , show that the acceleration of each particle has magnitude 0.98 m s^{-2} . (5)
- (d) When the particles have been moving for 0.5 seconds, they are at the same level.
 - (i) Find the speed of the particles at this time. (2)
 - (ii) Find d . (4)

(Total 14 marks)

Q20.

Two forces, $\mathbf{P} = (6\mathbf{i} - 3\mathbf{j})$ newtons and $\mathbf{Q} = (3\mathbf{i} + 15\mathbf{j})$ newtons, act on a particle. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular.

- (a) Find the resultant of $\mathbf{P}$ and $\mathbf{Q}$. (2)
- (b) Calculate the magnitude of the resultant of $\mathbf{P}$ and $\mathbf{Q}$. (2)
- (c) When these two forces act on the particle, it has an acceleration of $(1.5\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-2}$. Find the mass of the particle. (2)
- (d) The particle was initially at rest at the origin.
 - (i) Find an expression for the position vector of the particle when the forces have been applied to the particle for t seconds. (2)

- (ii) Find the distance of the particle from the origin when the forces have been applied to the particle for 2 seconds.

(2)

(Total 10 marks)

Q21.

A particle moves on a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

At time t seconds, the position vector of the particle is $\mathbf{r}$ metres, where

$$\mathbf{r} = \left(2e^{\frac{1}{2}t} - 8t + 5 \right) \mathbf{i} + (t^2 - 6t) \mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(3)

- (b) (i) Find the speed of the particle when $t = 3$.

(2)

- (ii) State the direction in which the particle is travelling when $t = 3$.

(1)

- (c) Find the acceleration of the particle when $t = 3$.

(3)

- (d) The mass of the particle is 7 kg.

Find the magnitude of the resultant force on the particle when $t = 3$.

(3)

(Total 12 marks)

Q22.

A stone, of mass 0.05 kg, is moving along the smooth horizontal floor of a tank, which is filled with oil. At time t , the stone has speed v . As the stone moves, it experiences a resistance force of magnitude $0.08 v^2$.

- (a) Show that

$$\frac{dv}{dt} = -1.6v^2$$

(2)

- (b) The initial speed of the stone is 3 m s^{-1} .

Show that

$$v = \frac{15}{5 + 24t}$$

(5)

(Total 7 marks)

Q23.

A motorcycle and rider, of total mass 300 kg, are accelerating in a straight line along a horizontal road at 2.2 m s^{-2} .

- (a) Show that the magnitude of the resultant force acting on the motorcycle is 660 N.

(1)

- (b) A forward driving force of P newtons together with a resistance force of magnitude 400 newtons act on the motorcycle. Find P .

(2)

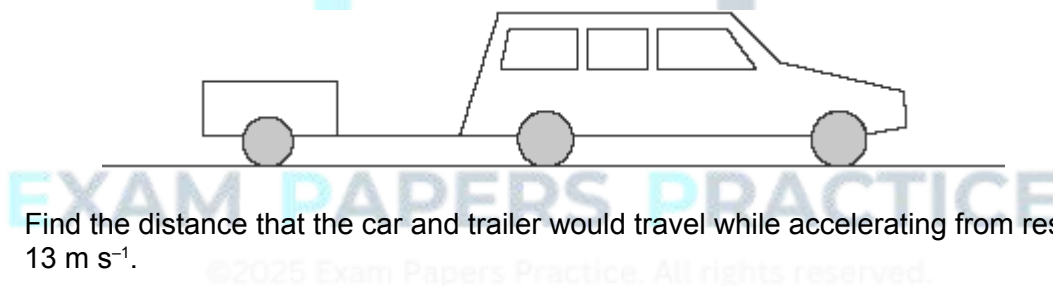
- (c) Find the time that it would take for the speed of the motorcycle to increase from 12 m s^{-1} to 23 m s^{-1} .

(3)

(Total 6 marks)

Q24.

A car, of mass 1400 kg, is towing a trailer, of mass 600 kg. The two vehicles accelerate together at 1.3 m s^{-2} along a straight horizontal road.



- (a) Find the distance that the car and trailer would travel while accelerating from rest to 13 m s^{-1} .

(3)

- (b) A forward driving force, of magnitude 3900 N, acts on the car. A resistance force, of magnitude 800 N, also acts on the car.

- (i) A resistance force, of magnitude P newtons, acts on the trailer. Find P .

(3)

- (ii) Find the magnitude of the force that the car exerts on the trailer.

(3)

(Total 9 marks)

Q25.

A car is travelling in a straight line on a horizontal road. A driving force, of magnitude 3000 N, acts in the direction of motion and a resistance force, of magnitude 600 N, opposes the motion of the car. Assume that no other horizontal forces act on the car.

- (a) Find the magnitude of the resultant force on the car. (2)
- (b) The mass of the car is 1200 kg. Find the acceleration of the car. (2)
- (Total 4 marks)**

Q26.

A block, of mass 14 kg, is held at rest on a rough horizontal surface. The coefficient of friction between the block and the surface is 0.25. A light inextensible string, which passes over a fixed smooth peg, is attached to the block. The other end of the string is attached to a particle, of mass 6 kg, which is hanging at rest.



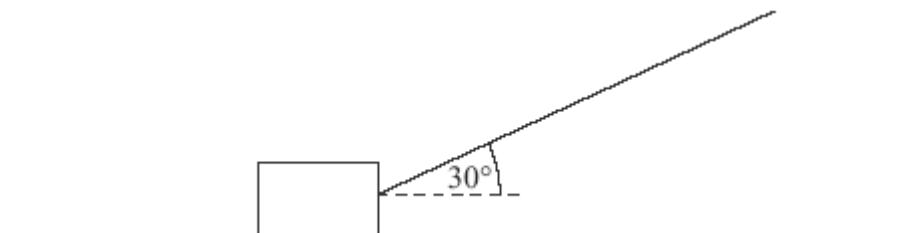
The block is released and begins to accelerate.

- (a) Find the magnitude of the friction force acting on the block. (3)
- (b) By forming two equations of motion, one for the block and one for the particle, show that the magnitude of the acceleration of the block and the particle is 1.225 m s^{-2} . (5)
- (c) Find the tension in the string. (2)
- (d) When the block is released, it is 0.8 metres from the peg. Find the speed of the block when it hits the peg. (3)
- (e) When the block reaches the peg, the string breaks and the particle falls a further 0.5 metres to the ground. Find the speed of the particle when it hits the ground. (3)

(Total 16 marks)

Q27.

The diagram shows a block, of mass 20 kg, being pulled along a rough horizontal surface by a rope inclined at an angle of 30° to the horizontal.



The coefficient of friction between the block and the surface is μ . Model the block as a particle which slides on the surface.

- (a) If the tension in the rope is 60 newtons, the block moves at a constant speed.
- (i) Show that the magnitude of the normal reaction force acting on the block is 166 N. (3)
- (ii) Find μ . (4)
- (b) If the rope remains at the same angle and the block accelerates at 0.8 m s^{-2} , find the tension in the rope. (5)
- (Total 12 marks)

Q28.

A particle moves under the action of a force, $\mathbf{F}$ newtons. At time t seconds, the velocity, $\mathbf{v} \text{ m s}^{-1}$, of the particle is given by

$$\mathbf{v} = (t^3 - 15t - 5)\mathbf{i} + (6t - t^2)\mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t . (3)
- (b) The mass of the particle is 4 kg.
- (i) Show that, at time t ,
- $$\mathbf{F} = (12t^2 - 60)\mathbf{i} + (24 - 8t)\mathbf{j}$$
- (2)
- (ii) Find the magnitude of $\mathbf{F}$ when $t = 2$. (4)
- (Total 9 marks)

Q29.

A stone, of mass m , is moving in a straight line along smooth horizontal ground.

At time t , the stone has speed v . As the stone moves, it experiences a total resistance

force of magnitude $\lambda m v^{\frac{3}{2}}$, where λ is a constant. No other horizontal force acts on the stone.

- (a) Show that

$$\frac{dv}{dt} = -\lambda v^{\frac{3}{2}}$$

(2)

- (b) The initial speed of the stone is 9 m s^{-1} .

Show that

$$v = \frac{36}{(2 + 3\lambda t)^2}$$

(7)

- (c) Find, in terms of λ , the time taken for the speed of the stone to drop to 4 m s^{-1} .

(3)

(Total 12 marks)

Q30.

A crane is used to lift a crate, of mass 70 kg , vertically upwards. As the crate is lifted, it accelerates uniformly from rest, rising 8 metres in 5 seconds .

- (a) Show that the acceleration of the crate is 0.64 m s^{-2} .

(2)

- (b) The crate is attached to the crane by a single cable. Assume that there is no resistance to the motion of the crate.

Find the tension in the cable.

(3)

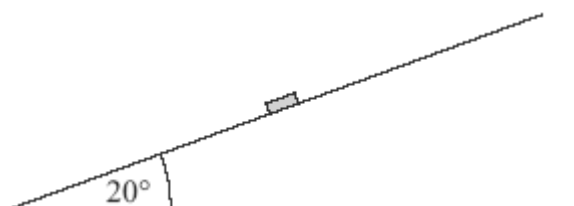
- (c) Calculate the average speed of the crate during these 5 seconds .

(1)

(Total 6 marks)

Q31.

A puck, of mass 0.2 kg , is placed on a slope inclined at 20° above the horizontal, as shown in the diagram.



The puck is hit so that initially it moves with a velocity of 4 m s^{-1} directly up the slope.

- (a) A simple model assumes that the surface of the slope is smooth.

- (i) Show that the acceleration of the puck up the slope is -3.35 m s^{-2} , correct to three significant figures.

(3)

- (ii) Find the distance that the puck will travel before it comes to rest.

(3)

- (iii) What will happen to the puck after it comes to rest?

Explain why.

(2)

- (b) A revised model assumes that the surface is rough and that the coefficient of friction between the puck and the surface is 0.5.

- (i) Show that the magnitude of the friction force acting on the puck during this motion is 0.921 N, correct to three significant figures.

(3)

- (ii) Find the acceleration of the puck up the slope.

(3)

- (iii) What will happen to the puck after it comes to rest in this case?

Explain why.

(2)

(Total 16 marks)

Q32.

A tractor, of mass 4000 kg, is used to pull a skip, of mass 1000 kg, over a rough horizontal surface. The tractor is connected to the skip by a rope, which remains taut and horizontal throughout the motion, as shown in the diagram.



Assume that only **two** horizontal forces act on the tractor. One is a driving force, which has magnitude P newtons and acts in the direction of motion. The other is the tension in the rope.

The coefficient of friction between the skip and the ground is 0.4.

The tractor and the skip accelerate at 0.8 m s^{-2} .

- (a) Show that the magnitude of the friction force acting on the skip is 3920 N.

(2)

- (b) Show that $P = 7920$.

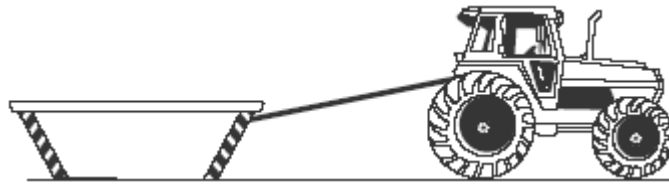
(3)

- (c) Find the tension in the rope.

(3)

- (d) Suppose that, during the motion, the rope is **not** horizontal, but inclined at a small

angle to the horizontal, with the higher end of the rope attached to the tractor, as shown in the diagram.



How would the magnitude of the friction force acting on the skip differ from that found in part (a)?

Explain why.

(2)

(Total 10 marks)

Q33.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the position vector, $\mathbf{r}$ metres, of the particle is given by

$$\mathbf{r} = (t^3 - 3t^2 + 4)\mathbf{i} + (4t + t^2)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(2)

- (b) The mass of the particle is 3 kg.

- (i) Find an expression for $\mathbf{F}$ at time t .

(3)

- (ii) Find the magnitude of $\mathbf{F}$ when $t = 3$.

(2)

- (c) Find the value of t when $\mathbf{F}$ acts due north.

(2)

(Total 9 marks)

Q34.

Two particles, A and B , are connected by a light inextensible string, which passes over a smooth peg. Particle A is on a rough horizontal surface and has mass 3 kg. Particle B hangs freely, as shown in the diagram, and has mass 2 kg. The coefficient of friction between A and the horizontal surface is μ .



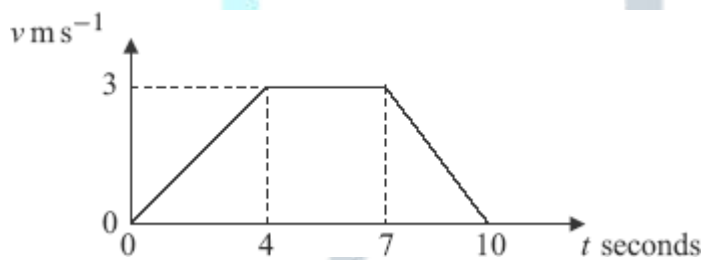
The particles are released from rest and move with a constant acceleration of magnitude 0.9 m s^{-2} .

- (a) Find the tension in the string. (3)
- (b) Draw and label a diagram to show the forces acting on particle A . (1)
- (c) Calculate the magnitude of the normal reaction force acting on A . (1)
- (d) Find the magnitude of the friction force that acts on A . (2)
- (e) Find μ . (2)

(Total 9 marks)

Q35.

The diagram shows a velocity–time graph for a lift.



- (a) Find the distance travelled by the lift. (3)
- (b) Find the acceleration of the lift during the first 4 seconds of the motion. (1)
- (c) The lift is raised by a single vertical cable. The mass of the lift is 400 kg. Find the tension in the cable during the first 4 seconds of the motion. (3)

(Total 7 marks)